Understanding and Upcoming Work

1.First, a neural network using 1×1 CNN layers was used to learn the SRFs. The main idea was to see whether the learned SRFs resemble the CIE/RGB filters and whether they can be used for classification tasks.

2.The initial experiments were performed on the Pavia University dataset using the visible range (400–700 nm). The learned SRFs were compared with the CIE filters, and the learned SRFs gave better classification performance.

3.The learned SRFs were also evaluated using SAM and compared with the CIE filters. From the results, the learned SRFs showed better performance than the CIE filters.

4.After that, the complete spectrum containing 103 bands was used, and only 3 primitives were considered. The goal was to check whether the model learns something similar to trichromatic color vision and how the learned SRFs compare with the CIE filters.

5.Then the focus shifted towards finding the optimal number of primitives. From what I understood, 3 primitives are not always sufficient, and the required number of primitives depends on the scene/material being analyzed. Accuracy tends to improve as the number of primitives increases.

6.Experiments were also performed with different numbers of primitives such as 3, 4, 7 and 10 to study how the SRFs change with the number of primitives and whether some of them become similar or redundant.

7.The Shadow class was also removed in one of the Pavia University experiments to study how the available classes and spectral information affect the learned primitives.

8.The same approach was later tested on the Indian Pines dataset and then on the Urban dataset containing materials such as grass, trees, roofs, bricks and roads to evaluate the behaviour of the learned SRFs on different scenes and materials.

9.I also understand that there are additional studies involving natural color composite generation, comparison with CIE curves, normalization methods, SRF initialization (CIE vs random initialization), and convergence behaviour.

10 Going forward, the next step is to replace the existing 1×1 CNN block with KAN and repeat the same set of experiments on Pavia University, Indian Pines and Urban datasets. The idea would be to compare CNN and KAN in terms of accuracy, learned SRFs, SAM analysis, optimal number of primitives, convergence behaviour, computational complexity and image quality measures.

WORKFLOW:

HSI Input
↓
KAN-BASED SRF Learning
↓
3/N Primitives
↓
Classification
↓
Accuracy + SAM + Analysis

Technologies that we are going to use:

Python, PyTorch, Jupyter Notebook running on the Welkin GPU environment, hyperspectral datasets (Pavia University, Indian Pines, and Urban), 1×1 CNN-based SRF learning (existing approach) , KAN (proposed approach), and SAM for evaluation and comparison.